

How can water reach a leaf above the soil?

Learning intention

Explain the root–stem–leaf water pathway and use dye evidence to identify internal transport in a cut shoot.

Trace a route.

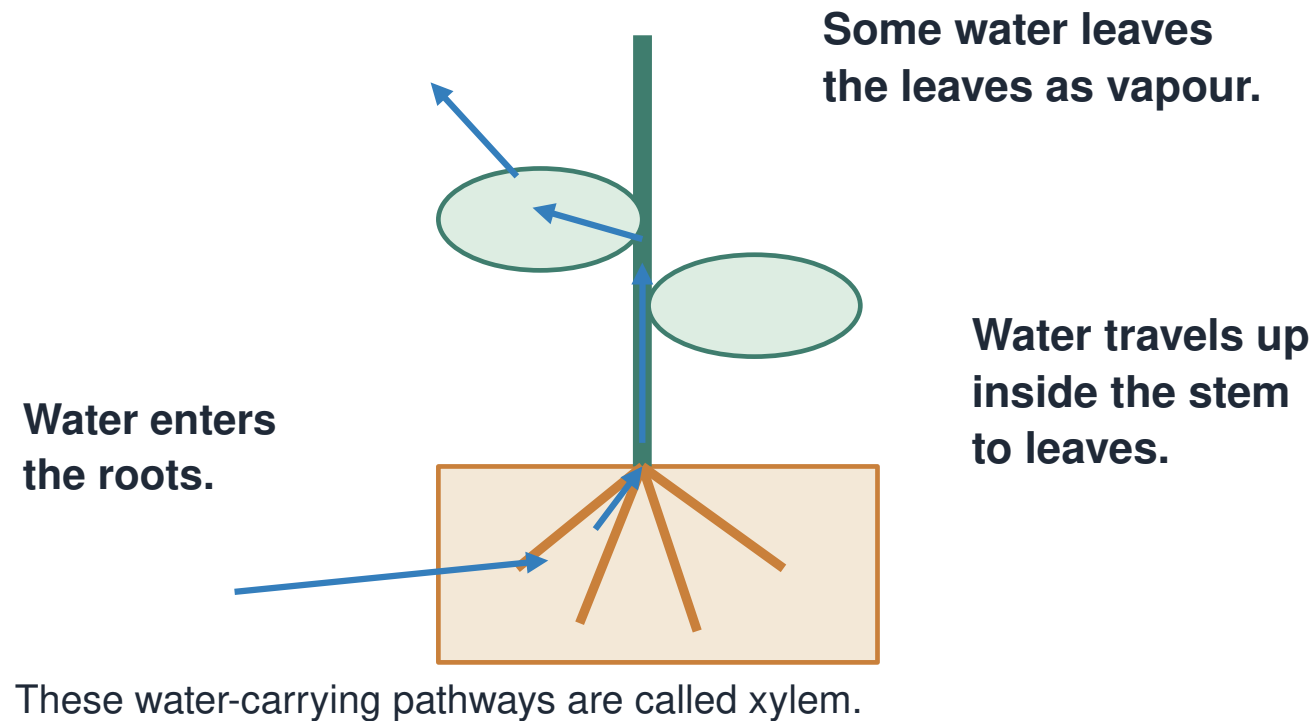
Use a marker to test part of it.

Retrieve the connected parts

1. Which part takes in soil water?
2. Which part connects roots and leaves?
3. Does water usually enter as solid soil?

I trace the rooted-plant route

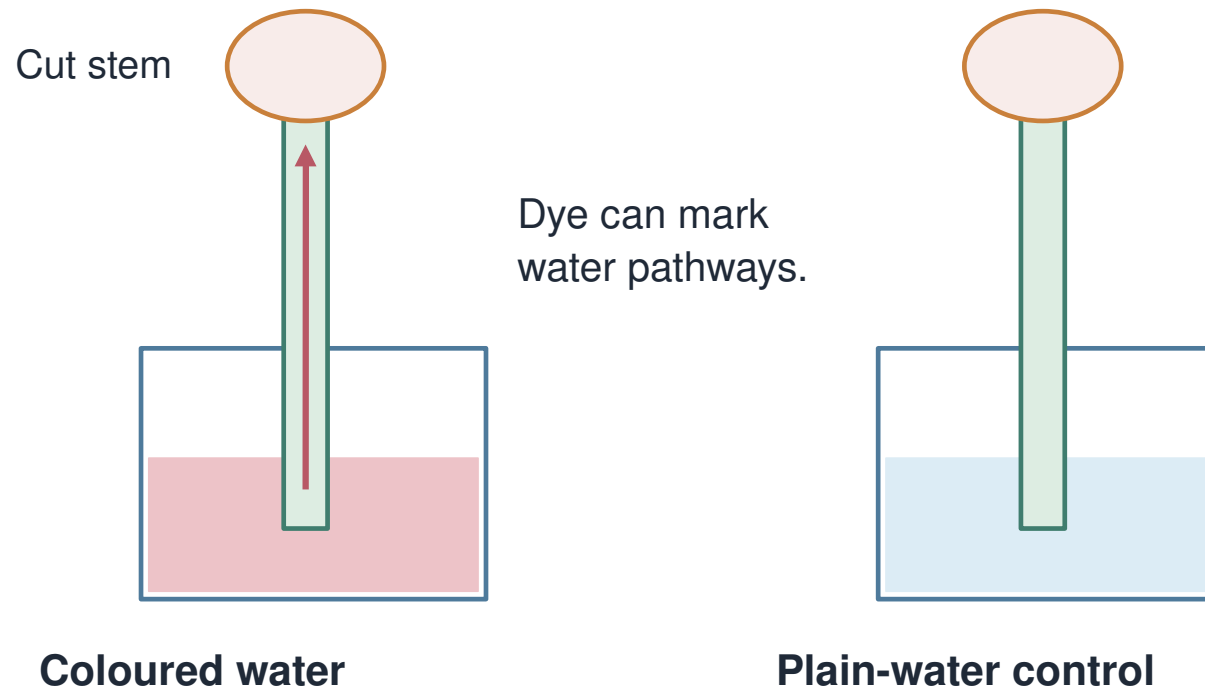
Rooted plant: water path



Soil → roots → stem → leaves. Water moves through internal pathways.

I mark water in a cut shoot

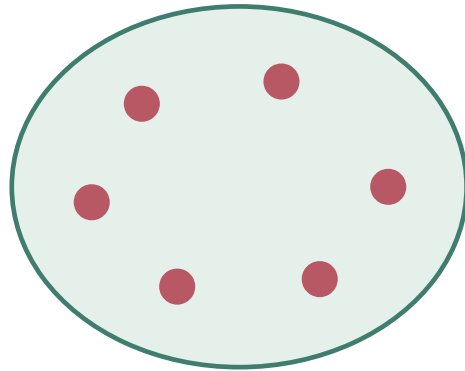
Cut shoots: this demonstration has no roots.



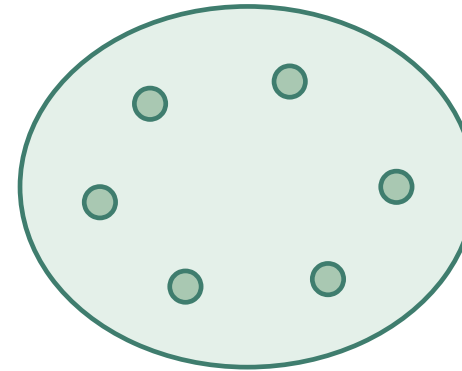
Colour helps us follow movement. A plain-water shoot is the comparison. This setup has no roots.

I inspect the interior evidence

Constructed observation: adult-cut sections



Dye shoot:
coloured internal paths



Plain-water shoot:
no dye-coloured paths

Colour appears in internal paths. Not all tissue is coloured equally.

Sequence the evidence before explaining

B: setup. C: stem colour later. D: petal colour later still.

Stage	Dye shoot	Plain-water comparison
Start	Placed in coloured water	Placed in plain water
Later	Internal red paths; then some red petals	No red dye colour

Constructed observations; timings vary in real demonstrations

Which claim reaches too far?

“Water moved within this cut shoot.”

“We proved how roots took up water.”

Use B to choose.

What does an internal red path support?

Focus on evidence inside the shoot.

- A** Coloured water moved through internal pathways.
- B** The demonstration directly observed root absorption.
- C** The dye is the plant's natural food.

Choose your science route

Complete the route and clues.

Write a transport explanation.

Challenge the overclaim.

Make your own evidence record

Use the shared evidence.

Explain one choice.

Check a label or number.

Check what the evidence can show

Point to the clue you used.

Remove an unsupported claim.

Lundy loop: improve the shared account

Choose how to share.

Hear one evidence-based challenge.

Improve or defend one idea.

Which limit made the explanation more trustworthy?

Keep the useful change

What changed in our account?

Which evidence helped?

Three final checks

- E1. Which part takes in water from soil?
- E2. Does water travel through internal stem pathways?
- E3. Did the cut-shoot experiment contain roots?